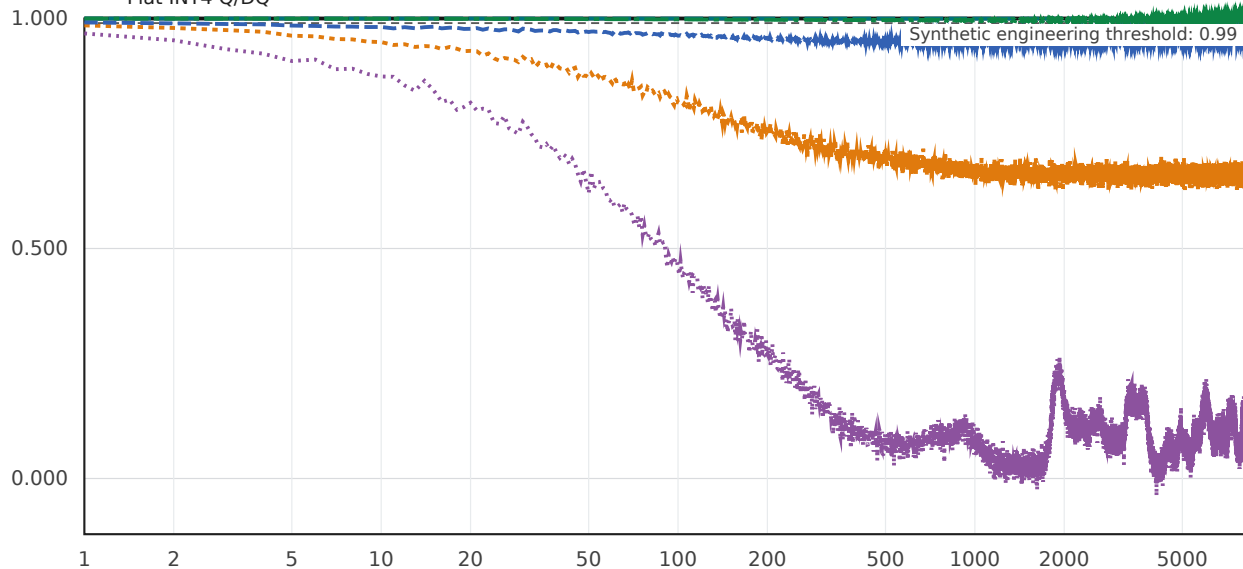


Output cosine similarity over decode tokens

— FP32 reference — BF16 operands/state (FP32 accum) — Native MXFP4 RS2/R3
- - - MXFP4 floating Q/DQ (act B32, state B32) - - - MXFP4 Q/DQ + MXFP8-E4M3 state (act B32, state B32)

... Flat INT4 Q/DQ



Decode token index (log scale)

Synthetic development high_retention; seed=64370; recurrence core; KxV state; 16 Q/K + 32 value heads; K=128, V=128; act/state B32/B32
Single deterministic trace (no uncertainty interval); relative L2 denominator=max(||reference||2, 1e-12); Floating Q/DQ baselines plus native encoded MXFP4